

PedraDB snapshot_backends — full ladder ×3 — b7fcde6

Harness: **snapshot_backends** (fjall / RocksDB / PedraDB). Config: VALUE=200 · SEQUENTIAL=1 · CACHE=unset (~1 GiB) · PEDRA_STAGE_MAX_BYTES=unset · CELLS=unset (full) · TMPDIR=/tmp · 1 backend/process. Scales: 1M, 10M, 25M, 100M, 250M, 500M × 3 runs × 3 backends = **54 cells**, all code=0. fjall settle at 500M SKIPPED (disk headroom). Campaign wall: 2026-10-03T03:59Z → 06:52Z.

1. Scoreboard

Scale	r1 fjall	r1 rocks	r1 pedra	r2 fjall	r2 rocks	r2 pedra	r3 fjall	r3 rocks	r3 pedra
1M	■	■	■	■	■	■	■	■	■
10M	■	■	■	■	■	■	■	■	■
25M	■	■	■	■	■	■	■	■	■
100M	■	■	■	■	■	■	■	■	■
250M	■	■	■	■	■	■	■	■	■
500M	■	■	■	■	■	■	■	■	■

2. Aggregates — median of r1/r2/r3 (code=0 only)

1M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	0.8s / 1.18 M/s → 0.32 GiB	0.6s / 1.62 M/s → 0.22 GiB	0.3s / 3.27 M/s → 0.20 GiB
Settle	0.0s → 0.33 GiB	0.8s → 0.21 GiB	0.1s → 0.24 GiB
Cold hit p50	2.90 μs	11.70 μs	12.90 μs
Cold miss p50	434.0 ns	579.0 ns	140.0 ns
probe_miss (crit)	276.3 ns	560.7 ns	110.7 ns
get_hit	1.78 μs	2.14 μs	2.23 μs
prefix_scan	203.93 μs	156.16 μs	125.87 μs
lookup_get_loop	167.16 μs	209.83 μs	213.80 μs
lookup_multi_get	178.46 μs	184.06 μs	221.12 μs

10M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	10.0s / 1.00 M/s → 2.41 GiB	6.0s / 1.67 M/s → 2.49 GiB	3.0s / 3.38 M/s → 2.35 GiB
Settle	6.7s → 4.36 GiB	3.1s → 2.10 GiB	0.2s → 2.40 GiB
Cold hit p50	9.60 μs	13.30 μs	14.90 μs
Cold miss p50	380.0 ns	304.0 ns	140.0 ns
probe_miss (crit)	336.2 ns	281.0 ns	110.6 ns

Metric	fjall	RocksDB	PedraDB
get_hit	7.52 μ s	6.52 μ s	8.70 μ s
prefix_scan	203.50 μ s	153.58 μ s	127.87 μ s
lookup_get_loop	745.23 μ s	664.37 μ s	856.21 μ s
lookup_multi_get	784.84 μ s	741.07 μ s	874.71 μ s

25M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	24.6s / 1.02 M/s $\rightarrow$ 5.46 GiB	14.7s / 1.70 M/s $\rightarrow$ 6.78 GiB	7.4s / 3.39 M/s $\rightarrow$ 5.94 GiB
Settle	18.6s $\rightarrow$ 9.92 GiB	4.1s $\rightarrow$ 5.24 GiB	0.2s $\rightarrow$ 5.99 GiB
Cold hit p50	18.10 μ s	10.30 μ s	15.60 μ s
Cold miss p50	390.0 ns	310.0 ns	140.0 ns
probe_miss (crit)	348.5 ns	286.1 ns	111.2 ns
get_hit	9.36 μ s	8.96 μ s	10.85 μ s
prefix_scan	204.38 μ s	153.70 μ s	133.34 μ s
lookup_get_loop	926.24 μ s	884.71 μ s	1.05 ms
lookup_multi_get	983.56 μ s	983.63 μ s	1.08 ms

100M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	101.6s / 0.98 M/s $\rightarrow$ 21.41 GiB	60.5s / 1.65 M/s $\rightarrow$ 24.28 GiB	29.8s / 3.35 M/s $\rightarrow$ 23.88 GiB
Settle	76.6s $\rightarrow$ 41.10 GiB	5.9s $\rightarrow$ 20.98 GiB	0.2s $\rightarrow$ 23.94 GiB
Cold hit p50	50.80 μ s	40.70 μ s	48.20 μ s
Cold miss p50	443.0 ns	333.0 ns	140.0 ns
probe_miss (crit)	396.3 ns	309.8 ns	110.4 ns
get_hit	27.18 μ s	34.27 μ s	31.33 μ s
prefix_scan	204.37 μ s	166.65 μ s	136.42 μ s
lookup_get_loop	2.55 ms	3.25 ms	3.25 ms
lookup_multi_get	2.78 ms	3.27 ms	3.45 ms

250M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	259.7s / 0.96 M/s $\rightarrow$ 52.79 GiB	147.0s / 1.70 M/s $\rightarrow$ 55.16 GiB	90.8s / 2.75 M/s $\rightarrow$ 59.79 GiB
Settle	200.8s $\rightarrow$ 102.00 GiB	5.7s $\rightarrow$ 52.45 GiB	0.6s $\rightarrow$ 59.85 GiB
Cold hit p50	77.90 μ s	44.40 μ s	54.90 μ s
Cold miss p50	419.0 ns	373.0 ns	141.0 ns
probe_miss (crit)	380.4 ns	347.8 ns	110.2 ns
get_hit	43.50 μ s	38.33 μ s	41.94 μ s
prefix_scan	204.82 μ s	169.98 μ s	177.39 μ s
lookup_get_loop	3.72 ms	3.81 ms	4.24 ms
lookup_multi_get	3.62 ms	4.14 ms	4.39 ms

500M — mediana

Metric	fjall	RocksDB	PedraDB
Hydrate	547.0s / 0.91 M/s → 105.31 GiB	296.3s / 1.69 M/s → 108.57 GiB	212.9s / 2.35 M/s → 119.65 GiB
Settle	SKIPPED (disk)	10.2s → 104.90 GiB	1.2s → 119.70 GiB
Cold hit p50	46.60 µs	65.10 µs	58.40 µs
Cold miss p50	436.0 ns	372.0 ns	141.0 ns
probe_miss (crit)	394.8 ns	349.6 ns	113.6 ns
get_hit	47.69 µs	238.91 µs	68.50 µs
prefix_scan	204.28 µs	170.48 µs	171.17 µs
lookup get_loop	5.92 ms	21.04 ms	7.04 ms
lookup multi_get	6.24 ms	22.13 ms	8.14 ms

3. Individual runs (all 54 cells)

Each table is one scale × one run. Columns = backends. Absolute times from criterion mid / one-shot hydrate.

1M — run r1

Metric	ijail	RocksDB	PedraDB
Hydrate	0.8s / 1.23 M/s → 0.32 GiB	0.6s / 1.62 M/s → 0.22 GiB	0.3s / 3.23 M/s → 0.20 GiB
Settle	0.0s → 0.33 GiB	0.9s → 0.21 GiB	0.1s → 0.24 GiB
Cold hit	p50=2.9µs p99=6.3µs	p50=19.6µs p99=27.6µs	p50=13.1µs p99=24.7µs
Cold miss	p50=433.0ns p99=472.0ns	p50=579.0ns p99=721.0ns	p50=140.0ns p99=172.0ns
probe_miss (crit)	278.58 ns	579.22 ns	110.41 ns
get_hit	1.7775 µs	2.1359 µs	2.2309 µs
prefix_scan	203.96 µs	155.50 µs	128.35 µs
lookup_get_loop	167.13 µs	210.41 µs	213.80 µs
lookup_multi_get	175.78 µs	181.75 µs	221.12 µs
exit	0	0	0

1M — run r2

Metric	ijail	RocksDB	PedraDB
Hydrate	0.8s / 1.18 M/s → 0.32 GiB	0.6s / 1.62 M/s → 0.22 GiB	0.3s / 3.27 M/s → 0.20 GiB
Settle	0.0s → 0.33 GiB	0.8s → 0.21 GiB	0.1s → 0.24 GiB
Cold hit	p50=2.8µs p99=6.0µs	p50=11.2µs p99=15.2µs	p50=12.8µs p99=24.3µs
Cold miss	p50=434.0ns p99=469.0ns	p50=594.0ns p99=702.0ns	p50=140.0ns p99=174.0ns
probe_miss (crit)	276.34 ns	560.71 ns	110.71 ns
get_hit	1.8075 µs	2.1919 µs	2.2097 µs
prefix_scan	202.87 µs	156.16 µs	125.87 µs
lookup_get_loop	172.49 µs	209.83 µs	220.20 µs
lookup_multi_get	179.66 µs	187.37 µs	221.08 µs
exit	0	0	0

1M — run r3

Metric	ijail	RocksDB	PedraDB
Hydrate	0.9s / 1.15 M/s → 0.32 GiB	0.6s / 1.65 M/s → 0.22 GiB	0.3s / 3.57 M/s → 0.20 GiB
Settle	0.0s → 0.33 GiB	0.8s → 0.21 GiB	0.1s → 0.24 GiB
Cold hit	p50=2.9µs p99=6.5µs	p50=11.7µs p99=16.2µs	p50=12.9µs p99=24.3µs
Cold miss	p50=438.0ns p99=894.0ns	p50=570.0ns p99=665.0ns	p50=139.0ns p99=173.0ns
probe_miss (crit)	272.88 ns	555.15 ns	112.78 ns
get_hit	1.7307 µs	2.1238 µs	2.4847 µs
prefix_scan	203.93 µs	156.29 µs	125.18 µs
lookup_get_loop	167.16 µs	208.23 µs	208.52 µs
lookup_multi_get	178.46 µs	184.06 µs	222.82 µs
exit	0	0	0

10M — run r1

Metric	fiell	RocksDB	PedraDB
Hydrate	10.7s / 0.94 M/s → 2.41 GiB	5.9s / 1.69 M/s → 2.49 GiB	2.9s / 3.40 M/s → 2.35 GiB
Settle	7.7s → 4.36 GiB	3.0s → 2.10 GiB	0.3s → 2.40 GiB
Cold hit	p50=9.6µs p99=17.5µs	p50=13.3µs p99=19.5µs	p50=14.8µs p99=27.6µs
Cold miss	p50=380.0ns p99=411.0ns	p50=304.0ns p99=349.0ns	p50=141.0ns p99=173.0ns
probe_miss (crit)	335.29 ns	281.03 ns	110.57 ns
get_hit	7.7855 µs	6.5249 µs	8.5937 µs
prefix_scan	202.51 µs	152.69 µs	127.87 µs
lookup_get_loop	764.64 µs	671.45 µs	856.21 µs
lookup_multi_get	795.46 µs	743.38 µs	874.71 µs
exit	0	0	0

10M — run r2

Metric	fiell	RocksDB	PedraDB
Hydrate	9.8s / 1.02 M/s → 2.41 GiB	6.0s / 1.67 M/s → 2.42 GiB	3.0s / 3.37 M/s → 2.35 GiB
Settle	6.6s → 4.36 GiB	3.1s → 2.10 GiB	0.2s → 2.40 GiB
Cold hit	p50=9.6µs p99=17.8µs	p50=13.2µs p99=17.5µs	p50=15.0µs p99=27.5µs
Cold miss	p50=376.0ns p99=407.0ns	p50=304.0ns p99=351.0ns	p50=140.0ns p99=173.0ns
probe_miss (crit)	336.21 ns	280.83 ns	110.57 ns
get_hit	7.5143 µs	6.5306 µs	8.7645 µs
prefix_scan	203.50 µs	153.58 µs	125.84 µs
lookup_get_loop	742.13 µs	643.45 µs	844.46 µs
lookup_multi_get	784.84 µs	741.07 µs	861.06 µs
exit	0	0	0

10M — run r3

Metric	fiell	RocksDB	PedraDB
Hydrate	10.0s / 1.00 M/s → 2.41 GiB	6.1s / 1.63 M/s → 2.52 GiB	3.0s / 3.38 M/s → 2.35 GiB
Settle	6.7s → 4.36 GiB	3.7s → 2.10 GiB	0.2s → 2.40 GiB
Cold hit	p50=9.7µs p99=18.2µs	p50=13.6µs p99=18.2µs	p50=14.9µs p99=27.3µs
Cold miss	p50=385.0ns p99=429.0ns	p50=308.0ns p99=355.0ns	p50=140.0ns p99=175.0ns
probe_miss (crit)	339.24 ns	281.15 ns	110.29 ns
get_hit	7.5249 µs	6.4550 µs	8.7027 µs
prefix_scan	204.69 µs	153.73 µs	131.02 µs
lookup_get_loop	745.23 µs	664.37 µs	907.40 µs
lookup_multi_get	784.69 µs	729.83 µs	888.15 µs
exit	0	0	0

25M — run r1

Metric	fjall	RocksDB	PedraDB
Hydrate	24.6s / 1.02 M/s → 5.46 GiB	14.7s / 1.70 M/s → 6.77 GiB	7.3s / 3.42 M/s → 5.94 GiB
Settle	19.5s → 9.92 GiB	4.1s → 5.24 GiB	0.2s → 5.99 GiB
Cold hit	p50=19.8μs p99=43.9μs	p50=10.4μs p99=15.2μs	p50=15.6μs p99=28.9μs
Cold miss	p50=390.0ns p99=423.0ns	p50=310.0ns p99=541.0ns	p50=140.0ns p99=174.0ns
probe_miss (crit)	344.52 ns	286.67 ns	110.50 ns
get_hit	9.3638 μs	8.8355 μs	10.817 μs
prefix_scan	206.05 μs	153.81 μs	135.59 μs
lookup_get_loop	926.03 μs	885.60 μs	1.0466 ms
lookup_multi_get	1.0545 ms	989.26 μs	1.0675 ms
exit	0	0	0

25M — run r2

Metric	fjall	RocksDB	PedraDB
Hydrate	24.8s / 1.01 M/s → 5.46 GiB	15.4s / 1.63 M/s → 6.78 GiB	7.4s / 3.39 M/s → 5.94 GiB
Settle	18.5s → 9.92 GiB	3.8s → 5.24 GiB	0.2s → 5.99 GiB
Cold hit	p50=18.1μs p99=37.4μs	p50=10.3μs p99=18.9μs	p50=15.6μs p99=28.2μs
Cold miss	p50=391.0ns p99=470.0ns	p50=312.0ns p99=353.0ns	p50=140.0ns p99=175.0ns
probe_miss (crit)	348.49 ns	286.10 ns	111.15 ns
get_hit	9.3697 μs	8.9609 μs	11.118 μs
prefix_scan	204.38 μs	152.05 μs	126.99 μs
lookup_get_loop	926.24 μs	884.71 μs	1.0439 ms
lookup_multi_get	983.56 μs	983.63 μs	1.0775 ms
exit	0	0	0

25M — run r3

Metric	fjall	RocksDB	PedraDB
Hydrate	24.5s / 1.02 M/s → 5.46 GiB	14.7s / 1.70 M/s → 7.37 GiB	7.5s / 3.34 M/s → 5.94 GiB
Settle	18.6s → 10.13 GiB	4.8s → 5.24 GiB	0.4s → 5.99 GiB
Cold hit	p50=11.2μs p99=19.5μs	p50=10.3μs p99=15.5μs	p50=16.9μs p99=34.3μs
Cold miss	p50=390.0ns p99=425.0ns	p50=310.0ns p99=342.0ns	p50=141.0ns p99=270.0ns
probe_miss (crit)	350.88 ns	284.86 ns	111.70 ns
get_hit	9.1726 μs	8.9691 μs	10.849 μs
prefix_scan	203.16 μs	153.70 μs	133.34 μs
lookup_get_loop	928.48 μs	858.35 μs	1.0890 ms
lookup_multi_get	973.38 μs	974.19 μs	1.0792 ms
exit	0	0	0

100M — run r1

Metric	fjall	RocksDB	PedraDB
Hydrate	104.0s / 0.96 M/s → 21.41 GiB	60.9s / 1.64 M/s → 24.28 GiB	30.1s / 3.33 M/s → 23.88 GiB
Settle	78.4s → 41.10 GiB	6.1s → 20.98 GiB	0.4s → 23.94 GiB

Metric	fjall	RocksDB	PedraDB
Cold hit	p50=50.8µs p99=98.3µs	p50=42.4µs p99=1.2ms	p50=39.3µs p99=245.6µs
Cold miss	p50=445.0ns p99=481.0ns	p50=333.0ns p99=381.0ns	p50=141.0ns p99=175.0ns
probe_miss (crit)	402.26 ns	309.82 ns	110.23 ns
get_hit	28.314 µs	85.341 µs	31.330 µs
prefix_scan	209.01 µs	166.01 µs	132.70 µs
lookup_get_loop	2.6234 ms	6.6052 ms	3.1342 ms
lookup_multi_get	3.0095 ms	6.3737 ms	16.981 ms
exit	0	0	0

100M — run r2

Metric	fjall	RocksDB	PedraDB
Hydrate	101.6s / 0.98 M/s → 21.41 GiB	59.4s / 1.68 M/s → 24.30 GiB	29.5s / 3.39 M/s → 23.88 GiB
Settle	76.4s → 41.10 GiB	5.9s → 20.98 GiB	0.2s → 23.94 GiB
Cold hit	p50=47.3µs p99=88.3µs	p50=40.7µs p99=54.9µs	p50=48.2µs p99=262.8µs
Cold miss	p50=443.0ns p99=478.0ns	p50=330.0ns p99=369.0ns	p50=139.0ns p99=176.0ns
probe_miss (crit)	396.27 ns	308.77 ns	110.44 ns
get_hit	27.184 µs	27.325 µs	28.431 µs
prefix_scan	204.37 µs	166.65 µs	136.42 µs
lookup_get_loop	2.4814 ms	2.6281 ms	3.2496 ms
lookup_multi_get	2.7804 ms	2.8327 ms	3.4463 ms
exit	0	0	0

100M — run r3

Metric	fjall	RocksDB	PedraDB
Hydrate	100.5s / 0.99 M/s → 21.41 GiB	60.5s / 1.65 M/s → 24.04 GiB	29.8s / 3.35 M/s → 23.88 GiB
Settle	76.6s → 41.10 GiB	5.8s → 20.98 GiB	0.2s → 23.94 GiB
Cold hit	p50=53.9µs p99=130.9µs	p50=33.5µs p99=526.3µs	p50=48.7µs p99=333.6µs
Cold miss	p50=423.0ns p99=463.0ns	p50=355.0ns p99=617.0ns	p50=140.0ns p99=173.0ns
probe_miss (crit)	386.18 ns	331.28 ns	111.14 ns
get_hit	26.373 µs	34.274 µs	31.998 µs
prefix_scan	203.13 µs	168.22 µs	136.76 µs
lookup_get_loop	2.5549 ms	3.2492 ms	3.3912 ms
lookup_multi_get	2.6694 ms	3.2736 ms	3.3785 ms
exit	0	0	0

250M — run r1

Metric	fjall	RocksDB	PedraDB
Hydrate	259.7s / 0.96 M/s → 52.79 GiB	146.7s / 1.70 M/s → 55.14 GiB	91.7s / 2.73 M/s → 59.79 GiB
Settle	207.6s → 102.00 GiB	5.7s → 52.45 GiB	0.7s → 59.85 GiB
Cold hit	p50=68.3µs p99=998.0µs	p50=38.8µs p99=53.6µs	p50=53.6µs p99=111.9µs
Cold miss	p50=463.0ns p99=499.0ns	p50=370.0ns p99=409.0ns	p50=141.0ns p99=174.0ns

Metric	fjall	RocksDB	PedraDB
probe_miss (crit)	415.18 ns	349.34 ns	110.46 ns
get_hit	46.467 µs	36.940 µs	45.203 µs
prefix_scan	204.82 µs	169.98 µs	177.39 µs
lookup_get_loop	4.3262 ms	3.8143 ms	4.2404 ms
lookup_multi_get	4.6502 ms	4.1354 ms	4.3876 ms
exit	0	0	0

250M — run r2

Metric	fjall	RocksDB	PedraDB
Hydrate	257.6s / 0.97 M/s → 52.83 GiB	147.0s / 1.70 M/s → 55.16 GiB	89.5s / 2.79 M/s → 59.79 GiB
Settle	195.9s → 102.00 GiB	4.3s → 52.45 GiB	0.6s → 59.85 GiB
Cold hit	p50=78.7µs p99=119.5µs	p50=44.4µs p99=58.9µs	p50=54.9µs p99=178.9µs
Cold miss	p50=413.0ns p99=451.0ns	p50=373.0ns p99=416.0ns	p50=139.0ns p99=192.0ns
probe_miss (crit)	371.57 ns	347.77 ns	110.24 ns
get_hit	35.678 µs	38.325 µs	41.025 µs
prefix_scan	204.80 µs	170.02 µs	178.33 µs
lookup_get_loop	3.0785 ms	3.4439 ms	3.9840 ms
lookup_multi_get	3.4211 ms	3.8029 ms	4.2989 ms
exit	0	0	0

250M — run r3

Metric	fjall	RocksDB	PedraDB
Hydrate	261.9s / 0.95 M/s → 52.62 GiB	147.3s / 1.70 M/s → 56.02 GiB	90.8s / 2.75 M/s → 59.79 GiB
Settle	200.8s → 102.00 GiB	6.4s → 52.45 GiB	0.6s → 59.85 GiB
Cold hit	p50=77.9µs p99=629.9µs	p50=45.6µs p99=443.3µs	p50=55.9µs p99=280.2µs
Cold miss	p50=419.0ns p99=455.0ns	p50=375.0ns p99=419.0ns	p50=142.0ns p99=204.0ns
probe_miss (crit)	380.41 ns	339.97 ns	109.83 ns
get_hit	43.505 µs	41.577 µs	41.937 µs
prefix_scan	206.43 µs	169.17 µs	174.33 µs
lookup_get_loop	3.7157 ms	4.2743 ms	4.7129 ms
lookup_multi_get	3.6185 ms	4.2771 ms	6.2413 ms
exit	0	0	0

500M — run r1

Metric	fjall	RocksDB	PedraDB
Hydrate	544.8s / 0.92 M/s → 105.31 GiB	296.3s / 1.69 M/s → 112.15 GiB	216.0s / 2.32 M/s → 119.65 GiB
Settle	SKIPPED (disk)	16.1s → 104.90 GiB	1.3s → 119.70 GiB
Cold hit	p50=204.1µs p99=1.6ms	p50=62.7µs p99=120.9µs	p50=58.1µs p99=728.4µs
Cold miss	p50=424.0ns p99=469.0ns	p50=372.0ns p99=411.0ns	p50=140.0ns p99=253.0ns
probe_miss (crit)	380.78 ns	352.44 ns	116.77 ns
get_hit	286.41 µs	50.625 µs	68.505 µs

Metric	fjall	RocksDB	PedraDB
prefix_scan	202.61 µs	170.52 µs	176.06 µs
lookup_get_loop	27.875 ms	4.3391 ms	7.2349 ms
lookup_multi_get	42.377 ms	4.5529 ms	7.9990 ms
exit	0	0	0

500M — run r2

Metric	fjall	RocksDB	PedraDB
Hydrate	547.9s / 0.91 M/s → 105.31 GiB	292.8s / 1.71 M/s → 108.57 GiB	211.8s / 2.36 M/s → 119.65 GiB
Settle	SKIPPED (disk)	10.2s → 104.90 GiB	1.2s → 119.70 GiB
Cold hit	p50=45.3µs p99=508.8µs	p50=65.1µs p99=1.7ms	p50=58.4µs p99=246.5µs
Cold miss	p50=439.0ns p99=474.0ns	p50=367.0ns p99=458.0ns	p50=141.0ns p99=175.0ns
probe_miss (crit)	403.80 ns	345.78 ns	110.26 ns
get_hit	42.730 µs	238.91 µs	49.818 µs
prefix_scan	204.76 µs	168.18 µs	171.17 µs
lookup_get_loop	5.9166 ms	21.038 ms	5.4920 ms
lookup_multi_get	5.8662 ms	22.128 ms	8.9897 ms
exit	0	0	0

500M — run r3

Metric	fjall	RocksDB	PedraDB
Hydrate	547.0s / 0.91 M/s → 105.31 GiB	302.5s / 1.65 M/s → 105.39 GiB	212.9s / 2.35 M/s → 119.65 GiB
Settle	SKIPPED (disk)	9.8s → 104.90 GiB	1.2s → 119.69 GiB
Cold hit	p50=46.6µs p99=554.8µs	p50=435.8µs p99=3.3ms	p50=58.4µs p99=237.0µs
Cold miss	p50=436.0ns p99=467.0ns	p50=380.0ns p99=464.0ns	p50=142.0ns p99=177.0ns
probe_miss (crit)	394.79 ns	349.57 ns	113.55 ns
get_hit	47.688 µs	581.93 µs	97.844 µs
prefix_scan	204.28 µs	170.48 µs	162.61 µs
lookup_get_loop	5.1628 ms	41.681 ms	7.0434 ms
lookup_multi_get	6.2417 ms	37.838 ms	8.1374 ms
exit	0	0	0

4. Hydrate rate matrix (M/s) — every run

Scale	r	fjall M/s	Rocks M/s	Pedra M/s	Pedra/Rocks
1M	r1	1.23	1.62	3.23	1.99×
1M	r2	1.18	1.62	3.27	2.02×
1M	r3	1.15	1.65	3.57	2.16×
10M	r1	0.94	1.69	3.40	2.01×
10M	r2	1.02	1.67	3.37	2.02×
10M	r3	1.00	1.63	3.38	2.07×
25M	r1	1.02	1.70	3.42	2.01×
25M	r2	1.01	1.63	3.39	2.08×
25M	r3	1.02	1.70	3.34	1.96×
100M	r1	0.96	1.64	3.33	2.03×
100M	r2	0.98	1.68	3.39	2.02×
100M	r3	0.99	1.65	3.35	2.03×
250M	r1	0.96	1.70	2.73	1.61×
250M	r2	0.97	1.70	2.79	1.64×
250M	r3	0.95	1.70	2.75	1.62×
500M	r1	0.92	1.69	2.32	1.37×
500M	r2	0.91	1.71	2.36	1.38×
500M	r3	0.91	1.65	2.35	1.42×

5. README / docs coverage — what this campaign re-ran

Re-ran (this PDF): sorted-ingest snapshot_backends at 1M / 10M / 25M / 100M / 250M / 500M × 3 runs × fjall/rocks/pedra (README sorted-ingest table + extension).

NOT re-run (still in README / docs/benchmarks.md):

Harness	Cell (README name)	docs id / notes
rocks-parity-bench	Overwrite, 25M keys, 4 clients	deps_cache_overwrite_mc4
rocks-parity-bench	Batched writes, 4 clients	deps_apply_batch_mc4
rocks-parity-bench	Read-heavy mix, 4 clients (95% reads)	ycsb_b_mc4
rocks-parity-bench	Lookup of a missing key, 100M keys	qs_neg_lookup (QPS; ≠ probe_miss)
rocks-parity-bench	Read-modify-write, 4 clients (open loss)	ycsb_f_mc4
rocks-parity-bench	prefix scan 100M @ 4 GiB bounded cache (open)	different cell from snapshot 1.05×

Harness	Cell (README name)	docs id / notes
rocks-parity-bench / fjall	Random read/write, 1M keys	ycsb_a_mc4 vs fjall
rocks-parity-bench / fjall	Scan, 1,024 keys	fjall scan cell
scale-parity-bench	Fjall column scale ladder 25M/100M	cross-harness fjall ratios

Config note vs official protocol (docs/benchmarks.md): official published legs use CACHE=256 MiB and PEDRA_STAGE_MAX_BYTES=64 MiB. This campaign used CACHE unset (~1 GiB) and STAGE unset (defaults of the current scale-bench campaign).